

AI Methodology Appendix

P.R.I.M.E. Entry 1: Code Generation

Prompt: I asked the AI assistant to help generate Python code for data loading, EDA, balance checks, OLS, Double Machine Learning, visualization, and Streamlit dashboard preparation.

Response: The assistant produced structured Colab-compatible code.

Iterate: I adjusted the code to match the assignment requirements and available dataset.

Modify: I verified the imports, variable names, and output files.

Evaluate: I ran the notebook and checked that the code produced interpretable outputs.

P.R.I.M.E. Entry 2: Analysis Assistance

Prompt: I asked the AI assistant to explain the difference between prediction and causation and to help frame the identification strategy.

Response: The assistant explained why a prediction model is insufficient for estimating the causal effect of job training.

Iterate: I connected the explanation to the assignment's requirement for an explicit identification defense.

Modify: I revised the language to focus on conditional independence, selection bias, and the DML estimate of \$839.

Evaluate: I checked that the final explanation matched the empirical design and did not overclaim beyond the identification assumption.

P.R.I.M.E. Entry 3: Writing Support

Prompt: I asked the AI assistant to help draft the executive summary, technical report, threats memo, and AI appendix.

Response: The assistant produced structured writing using the required project components.

Iterate: I reviewed the drafts for consistency with my analysis results.

Modify: I edited wording, checked assumptions, and ensured that the reports did not confuse prediction with causation.

Evaluate: I confirmed that the final documents addressed the rubric requirements.

Verification Statement

All AI-generated material was reviewed and modified by the student. Code outputs were checked by running the notebook. The final numerical results were inserted automatically from the executed analysis: DML estimate \$839, 95 percent confidence interval [\$-556, \$2,235]. Causal claims were evaluated against the conditional independence assumption.